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Sex differences in Cognitive Abilities Test scores: A UK national picture

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Background and aims. There is uncertainty about the extent or even existence of sex differences in the mean and variability of reasoning test scores (Jensen, 1998; Lynn, 1994, 1998; Mackintosh, 1996). This paper analyses the Cognitive Abilities Test (CAT) scores of a large and representative sample of UK pupils to determine the extent of any sex differences.

Sample. A nationally representative UK sample of over 320,000 school pupils aged 11–12 years was assessed on the CAT (third edition) between September 2001 and August 2003. The CAT includes separate nationally standardized tests for verbal, quantitative, and non-verbal reasoning. The size and recency of the sample is unprecedented in research on this issue.

Methods. The sheer size of the sample ensures that any sex difference will achieve statistical significance. Therefore, effect sizes (*d*) and variance ratios (VR) are employed to evaluate the magnitude of sex differences in mean scores and in score variability, respectively.

Results. The mean verbal reasoning score for girls was 2.2 standard score points higher than the mean for boys, but only 0.3 standard points in favour of girls for non-verbal reasoning (NVR), and 0.7 points in favour of boys for quantitative reasoning (QR). However, for all three tests there were substantial sex differences in the standard deviation of scores, with greater variance among boys. Boys were over represented relative to girls at both the top and the bottom extremes for all tests, with the exception of the top 10% in verbal reasoning.

Conclusions. Given the small differences in means, explanations for sex differences in wider domains such examination attainment at age 16 need to look beyond conceptions of 'ability'. Boys tend to be both the lowest and the highest performers in terms of their reasoning abilities, which warns against the danger of stereotyping boys as low achievers.

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The question of sex differences in cognitive performance has a long history in psychology and education. The issue has a high profile within the current UK educational context. National testing in England has provided data to show that girls outperform boys in assessments of English at age 7, 11 and 14, although differences in mathematics and science are less clear cut. In public examinations at age 16, girls again achieve greater success than boys. For example, in General Certificate of Secondary Education (GCSE) public examinations in England in 2002, Department for Education and Skills (2002) statistics show that 57% of girls, but only 46% of boys, achieved five or more higher grade (A* to C) passes. In individual subjects, the proportion of girls achieving A* to C grades exceeded the proportion for boys not just in English but also in subjects where males have traditionally been thought to have an advantage, such as mathematics, business studies, design and technology, and science and information technology. The only GCSE subject in which the performance of boys exceeded that of girls was physics, in which 90% of boys, compared with 89% of girls, achieved an A* to C grade.

The public and media are intensely interested in this so-called 'gender gap', reflected in headlines such as 'Failing boys "public burden number one"' (1998); 'Gender gap widens to a gulf' (1999); 'Bright girls leave boys out-classed' (2000); 'Boys in crisis' (2000); 'The trouble with boys' (2000); 'GCSE gender gap continues to grow' (2002). This concern is not limited to the media. For example, Chris Woodhead, the former Chief Inspector of schools in England, described under-achieving boys as 'one of the most disturbing problems facing the education system' ('Failing boys,', 1998). There has also been a strong political input, involving national strategies, task groups, and targets. For example, one of the Welsh National Assembly's targets was that, by the year 2002, the under-achievement of boys against girls in national tests and examinations should be cut by 50% as compared with 1996.

We can locate this concern with the 'gender gap' within a long history of investigating sex differences in intellectual abilities. Does the gender gap in examination attainment reflect sex differences in more fundamental cognitive domains such as aspects of psychometric intelligence or reasoning abilities? Do boys and girls differ in their scores on IQ-type or reasoning abilities tests?

Sex differences in IQ

Early standardizations of the Stanford-Binet and Weschler-Bellevue IQ tests tended to indicate a small score difference favouring females, although these were not considered significant (Mackintosh, 1996, pp. 182–199). However, standardizations of the revised editions of the Wechsler intelligence scale for children (WISC-R) and Wechsler adult intelligence scale (WAIS-R) in the early 1980s showed a small difference, favouring males, of around 1.7 points on the WISC-R and 2.2 points on WAIS-R (Jensen & Reynolds, 1983; Reynolds, Chastain, Kaufman, & McClean, 1987). The results obtained on recent large, representative population samples are also equivocal. Thus, Hernstein and Murray (1994), describing the USA's National Longitudinal Survey of Youth 1979, described the tests scores of some 12,000 teenagers and young adults, and found a difference of 0.9 IQ points in favour of men. However, Lubinski and Humphreys (1990) analysed the test scores of 100,000 16-year-old US school students and found a difference of 0.3 IQ points in favour of girls.

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There continues to be debate on the extent, or even existence, of sex differences in the mean level of IQ scores (Colom, Juan-Espinosa, Abad, & Garcia, 2000; Halpern & LaMay, 2000; Jensen, 1998; Lynn, 1994, 1998; Lynn, Allik, & Irwing, 2004, Lynn & Irwing, 2004; Mackintosh, 1996). However, it is apparent in the majority of studies that, even when sex differences in mean IQ scores are found, they tend to be small. Intelligence is not a single homogeneous ability (Carroll, 1993) and IQ tests reflect this. Males tend to perform better on some subtests, and females on others; when these results are averaged across subtests, these differences tend cancel each other out. The main evidence for sex differences tends to come from differential performance in specific abilities.

Sex differences in specific abilities

Maccoby and Jacklin (1974) reviewed studies of sex differences published in American journals in the 10 years preceding 1974. They concluded that the sexes did not differ consistently in tests of composite abilities such as IQ. However, from adolescence onwards, there was evidence of girls' superiority in a variety of verbal abilities, which continued into adulthood. In contrast, there seemed to be a consistent trend for a male advantage from age 13 onwards in quantitative and visuospatial abilities.

Maccoby and Jacklin's book generated considerable debate (see Caplan, 1979) and their overall conclusions have been supported in some subsequent research (e.g. Feingold, 1992; Halpern, 1992; Halpern & LaMay, 2000) but not in others. For example, Hyde and Linn (1988) performed a meta-analysis of 165 studies of sex differences in verbal ability. They summarize their results using effect size (d) estimates, which are the difference between the mean scores for boys and girls divided by the pooled standard deviation. They concluded that there was a modest female superiority of $d = 0.20$ on tests of general verbal ability, $d = 0.22$ on anagrams, and $d = 0.33$ for speech production although, paradoxically, they also concluded there was a male advantage of $d = 0.16$ on verbal analogies, giving an overall verbal effect size of $d = 0.11$ in favour of girls, which they considered insubstantial. Similarly, for mathematical ability, Hyde, Fennema, and Lamon (1990) performed a meta-analysis of 100 studies and reported an overall effect size of only $d = 0.05$, and in favour of females. However, the results suggested significant interactions between student age, type of ability, and the selectivity of the sample. Thus differences favouring males tended to be restricted to the area of problem solving, emerged only at high school age (15–17 years), and were largest for self-selected samples, such as the US Scholastic Aptitude Test-Maths (SAT-Maths) compared with general population samples.

There is considerable variability in the outcomes of the many small studies included within the Hyde and Linn (1988) and Hyde *et al.* (1990) meta-analytic reviews. Perhaps the most compelling evidence in relation to sex differences will be found in the analysis of norms from standardized tests, where the sample is large and nationally representative on key demographic, educational and other relevant criteria. Two studies are particularly eminent in meeting these criteria. Feingold (1992) reviewed test norming statistics for four standardizations of the Differential Aptitude Test (DAT) between 1947 and 1980 with US students aged between 14 and 17 and above. The results summarized in Table 1 do not reveal the substantial male advantage in numerical ability, or the female advantage in verbal reasoning, that might be expected from Maccoby and Jacklin's (1974) conclusions, although the female advantage for language and spelling and the male advantage for spatial relations are more congruent. A paper by Hedges and Nowell (1995) is also particularly robust in terms of sample size

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Table 1. Gender differences in mean score and score variability averaged over four standardizations of the Differential Aptitude Test (DAT) between 1947 and 1980 with students aged between 14 and 18 and above (Feingold, 1992)

DAT battery	Effect size (*d*)	Variance ratio (VR)
Numerical ability	.05	1.11
Mechanical reasoning	.98	1.28
Space relations	.24	1.21
Spelling	−.50	1.12
Verbal reasoning	.05	0.96
Abstract reasoning	.08	1.01
Language	−.43	0.99
Clerical speed and accuracy	−.03	0.94

and representativeness. They performed a secondary analysis of six large US national datasets collected between 1960 and 1992. The datasets involved people from age 15 to early 20s and all were based on large national probability samples. They concluded that females exhibited a slight tendency to perform better on tests of reading comprehension, perceptual speed, and associative memory, and males tended to perform better on tests of mathematics and social studies. However, with the exception of the male advantage on the vocational aptitude scales, the effect sizes were relatively small, less than $d = 0.2$.

Sex differences in variability in test scores

The majority of studies have only considered sex differences in mean scores. However, in an often-overlooked aspect of their review, Maccoby and Jacklin (1974) also concluded that males were more variable than females in mathematical and spatial abilities, although the sexes were equally variable in verbal ability. The issue of increased cognitive variability in males was previously discussed in detail by Heim (1970). Feingold (1992) analysed the results for the national standardizations of the DAT, the SAT, the WAIS and the California achievement tests. Males tended to be more variable than females in general knowledge, mechanical reasoning quantitative ability, spatial visualisation, and spelling. There was little difference in variability for most verbal tests, short-term memory, non-verbal reasoning and perceptual speed (see Table 1 for DAT results). Hedges and Nowell (1995) reported that males had greater variance than females in all but two of the areas they considered, typically in the order of 3–15% greater variability in boys' scores than in girls' scores. Cole (1997) also reported greater variability in boys' scores on many of the tests analysed. For example, at age 17, males outnumbered females in the top 10% on maths tests by 1.5–1, and in science by 2–1.

Sex differences in spread or variability are important because they help to explain why males may outnumber females among the highest scoring individuals in tests that show only a small male advantage in mean score (Feingold, 1992; Hedges & Nowell, 1995; Nowell & Hedges, 1998). The reverse was also true; in Hedges and Nowell's study, boys outnumbered girls in the bottom 10% for those tests with only a small female advantage in mean score (e.g. reading comprehension, perceptual speed and associative memory).

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Trends over time

In their meta-analysis of sex differences in verbal ability, Hyde and Linn (1988) reported a mean effect size of $d = 0.23$ (favouring girls) for studies conducted before 1973, but a mean effect size in the same direction of only $d = 0.10$ for studies completed from 1973 onwards. Similarly, for mathematics, Hyde *et al.* (1990) reported a mean effect size for studies published prior to 1973 of $d = 0.31$ (favouring boys), but only $d = 0.14$ in the same direction for the studies completed from 1974 onwards.

Other studies relate to attainment rather than reasoning tests, but suggest a similar trend. Nowell and Hedges (1998) based their assessment of time trends on the US National Assessment of Educational Progress (NAEP) data for 17-year-old students, which consists of tests of reading, writing, mathematics and science, similar in its curriculum focus to the England National Curriculum testing programme. They suggested that, over the period between 1971 and 1994, the small sex differences favouring males in mathematics and science scores appeared to have narrowed slightly, but that the relatively large sex differences favouring girls in reading and writing had not. Cole (1997) also reported an analysis of a nationally representative sample of US 15-year-olds (Project Talent) revealing an effect size for science that reduced the male advantage from about $d = 0.60$ to under $d = 0.20$ from 1960 to 1990, with mathematics showing a similar reduction from $d = 0.45$ to $d = 0.10$. However, females sustained their advantage in writing from 1960 to 1990 at approximately $d = 0.40$.

Studies in the UK

It is interesting that the large meta-analyses undertaken by Hyde and Linn (1988) and Hyde *et al.* (1990) specifically excluded all studies from outside the US. UK and other national studies are important because results found in the US are not consistently replicated in other countries (Feingold, 1994). However, very few studies have been completed in the UK that meet the stringent methodological criteria of large and nationally representative samples. Deary, Thorpe, Wilson, Starr, and Whalley (2003) reported an analysis of the Moray House Verbal Reasoning test completed at age 11 by almost all Scottish schoolchildren born in 1921 as part of the 1932 Scottish Mental Survey ($N = 87{,}498$). This is probably the only near-complete national examination of a whole year-of-birth cohort. Despite there being about 40,000 boys and girls, they found no sex difference in mean IQ score. However, there was greater variability among boys' scores, such that boys were overrepresented relative to girls at both the highest and lowest extremes.

The present study

It is important that the abilities assessed in studies are clearly defined. For example, Hyde and Linn (1988) note in their meta-analysis that, 'verbal ability' has been used as a category to include everything from quality of speech in 2-year-olds, to performance on the Peabody Picture Vocabulary Test (PPVT) at age 5 years, to essay writing by high school students, to solutions to anagrams and analogies'. Similarly, 'mathematical ability' has referred to varied measures such as computation, concepts or problem solving (Cole, 1997; Hyde *et al.*, 1990). Many of the measures reported by Hedges and Nowell (1995) do not focus on reasoning abilities at all, but rather on vocational aptitude (mechanical reasoning, electronic information, and auto and shop information) or school subjects such as science, mathematics and social studies. Performance in these areas might be strongly affected by differential male–female educational experience such as different subject choices and by differential drop-out from school after

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the compulsory years, particularly for the older students (aged over 16) who form the majority of the populations in their study.

The present study reports results from a large and representative UK national sample of 11- to 12-year-olds using the Cognitive Abilities Test (CAT) to address questions of sex differences in specific cognitive abilities scores in the UK. The study has several strengths in relation to previous reports. It focuses on the UK, in contrast to the majority of research that has been conducted within the US. It analyses the results for an extremely large and nationally representative sample of over 320,000 schoolchildren. It focuses on reasoning abilities rather than educational attainment in school subjects or vocational aptitude tests. Reasoning tests should be less affected by subject choices or by differential educational experiences than curriculum-related or vocational tests. It disaggregates verbal, quantitative, and non-verbal reasoning scores to allow a more sophisticated analysis of differences in abilities, in contrast to previous UK studies focusing only on overall IQ (e.g. Deary *et al.*, 2003). It focuses on early secondary school (ages 11–12), where all schoolchildren are in compulsory education, and the effects of selective drop-out – from education as a whole and from specific subjects – are removed. It reports recent results from tests completed in 2002 and 2003.

Method

Participants: The CAT3 data sample

The total dataset consisted of over 500,000 UK schoolchildren who completed CAT (third edition; CAT3) between September 2001 and August 2003. The largest proportion of schoolchildren completed level D, designed for 11- to 12-year-olds in the first year of secondary school. Level D scores were available for over 324,000 schoolchildren from 1,305 schools. The average age of schoolchildren completing Level D was 11 years and 7 months with a standard deviation 4.4 months (these means and *SD*s were identical for boys and for girls). Two-thirds of schoolchildren were in the age range 11.03–12.00. Within the total sample, 49.9% of schoolchildren were boys and 50.1% were girls, equivalent to the 2001 England average for the 11–12 year age range.

The sample of 320,000 participants represents almost half of the UK's population of 11- to 12-year-olds (approximately 700,000 children). However, sheer size does not of itself ensure the absence of selective bias in the sample. Over 84% of the schoolchildren taking CAT3 Level D were drawn from maintained, mainstream secondary schools in England. This subsample was compared with national statistics computed for all maintained, mainstream secondary schools in England on a dataset collected by the Department for Education and Skills in January 2001. The national dataset was analysed in relation to the selective status of the school (in areas of the country operating selection, grammar schools select the most able pupils and the rest attend secondary modern schools; in non-selective areas, all children attended comprehensive schools). In addition, five bands, each containing 20% of all schools nationally, were created to describe the range of school variation on a number of key variables, including entitlement to free school meals (an indication of the economic disadvantage of the school population), the proportion of schoolchildren from ethnic minority groups, and the proportion of schoolchildren with English as an additional language. The results are presented in Table 2.

The CAT sample of 1,046 schools includes almost one third (30%) of all maintained, mainstream secondary schools in England. This figure substantially underestimates the total proportion of schools using CAT, because a large minority of English

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Table 2. Comparison of sample against averages for all maintained mainstream secondary schools in England

		All maintained mainstream secondary schools in England January 2001[a]		Maintained mainstream secondary schools in England using CAT3 with Y7[b]	
Number of schools		3,481		1,046	30%
Selective status	Comprehensive	3,140	90.2%	941	90.0%
	Secondary modern	145	4.2%	53	5.1%
	Selective grammar	159	4.6%	41	3.9%
	Other	37	1.1%	11	1.1%
Entitlement to free school meals	Bottom 20%	710	20.4%	179	17.1%
	Low–middle 20%	697	20.0%	209	20.0%
	Middle 20%	682	19.6%	191	18.3%
	Middle–high 20%	696	20.0%	220	21.0%
	Top 20%	696	20.0%	247	23.6%
Ethnicity	Bottom 20%	740	21.3%	230	22.1%
	Low–middle 20%	663	19.0%	184	17.6%
	Middle 20%	692	19.9%	191	18.3%
	Middle–high 20%	684	19.6%	186	17.8%
	Top 20%	693	19.9%	252	24.2%
English as additional language	Bottom 20%	732	21.1%	232	22.2%
	Low–middle 20%	674	19.4%	194	18.5%
	Middle 20%	688	19.8%	179	17.1%
	Middle–high 20%	692	19.9%	199	19.0%
	Top 20%	695	20.0%	242	23.1%

[a] Includes middle schools deemed secondary schools.
[b] For the purpose of this analysis, CAT results from 292 schools in Scotland, Wales and Northern Ireland, and independent and special schools in England, have been excluded.

maintained secondary schools were still using CAT second edition (CAT2E) during this time period. In comparing the sample against national averages, the sample is broadly representative of all such schools in England. There are only slight variations from the national proportions so that, in terms of selective status, the proportion of schoolchildren entitled to free school meals, the proportion of ethnic minority schoolchildren, and the proportion of schoolchildren with English as an additional language, the sample is broadly representative of all schools nationally.

Cognitive Abilities Test (CAT3)

CAT3 is the most recent UK version of the CAT, and was published in July 2001 (Lohman *et al.*, 2001). The CAT provides an assessment of a child's reasoning abilities in the verbal reasoning (VR), quantitative reasoning (QR) and non-verbal reasoning (NVR) domains. Each domain is assessed by a separate battery of three tests. A child's mean score over the three batteries (mean CAT score) is also calculated. The test is divided into eight levels coded as Levels A to H, and is standardized in the UK across the age range 7;6–17;0 and above.

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We next describe the tests within each battery, giving indicative items.

CAT3 verbal reasoning (VR) battery

- *Verbal classification*. Given three words belonging to one class, select which further word from a list of five belongs to the same class (e.g. eye, ear, mouth: nose, smell, head, boy, speak).
- *Sentence completion*. Selecting one word from a list of five (e.g. John likes to ___ a football match: eat, help, watch, read, talk).
- *Verbal analogies*. Given one pair of words, complete a second pair from five possibilities (e.g. big → large; little → ?: boy, small, late, lively, more).

CAT3 quantitative reasoning (QR) battery

- *Number analogies*. Determine the relationship between the numbers in two example pairs and decide which of five options would complete a third pair in the same way (e.g. [9 → 3] [12 → 4] [27 → ?]: 5, 9, 13, 19, 21).
- *Number series*. Select one from five possible choices to complete the series (e.g. 2, 4, 6, 8, → ?: 9, 10, 11, 12, 13).
- *Equation building*. Select the one answer choice that can be calculated by combining all the given elements to create a valid equation (e.g. 2 2 3 + x: 6, 8, 9, 10, 11).

CAT3 non-verbal reasoning (NVR) battery

- *Figure classification*. Given three shapes belonging to one class, select which further shape from five alternatives belongs to the same class.
- *Figure analogies*. Given one pair of shapes, complete a second pair from five possibilities.
- *Figure analysis*. A piece of paper is folded and holes are punched through the paper. How will the paper look when it is unfolded?

These subtests include item types with a long pedigree (such as classification, analogies, and series) as well as relatively more recent forms (such as equation building and figure analysis). The resulting CAT battery scores have extremely high reliability (Strand, 2004) and strong validity correlations with later educational attainment (Smith, Fernandes, & Strand, 2001; Strand, 2003, 2006).

Results

Table 3 presents, for boys and girls separately, the mean score, standard deviation and sample size for standard age scores on each of the three batteries and the mean CAT score. All sex comparisons are highly statistically significant; girls had a higher mean score than boys on the verbal battery, the non-verbal battery and mean CAT score, and boys had a significantly higher mean score than girls on the quantitative battery. Boys had significantly greater variance on all four CAT measures. However, with samples of this size, even very small absolute differences are probably statistically significant at conventional p values. We therefore need to examine the magnitude of the sex differences in mean scores and in score variability by considering the effect size (d) and the variance ratio (VR).

Sex differences in mean scores

The effect size is the difference between the mean scores for boy and girls divided by the pooled standard deviation. Thus, the sex difference in mean verbal reasoning score of

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Table 3. Mean standard age score, standard deviation and sample size for boys and girls on CAT3 Level D with statistical significance, effect size, variance ratios and tail proportion ratios for each CAT battery

							Tail proportion ratios			
CAT battery	Statistic	Boys	Girls	Significance	Effect size (*d*)	Variance ratio (VR)	Lowest 5%	Lowest 10%	Top 10%	Top 5%
Verbal	Mean	98.4	100.6	$p < .0001$	0.15	1.09	1.53	1.42	0.86	0.86
	SD	15.1	14.5	$p < .0001$						
	N	158,093	158,457							
Quantitative	Mean	99.4	98.9	$p < .0001$	− 0.03	1.18	1.30	1.19	1.34	1.46
	SD	15.0	13.8	$p < .0001$						
	N	157,862	158,406							
Non-verbal	Mean	99.7	100.2	$p < .0001$	0.03	1.13	1.36	1.23	1.09	1.17
	SD	14.8	13.9	$p < .0001$						
	N	157,830	158,299							
Mean CAT	Mean	99.1	99.9	$p < .0001$	0.05	1.13	1.38	1.29	1.07	1.10
	SD	13.5	12.7	$p < .0001$						
	N	156,556	157,258							

Note. Positive effect size (*d*) indicates the female mean greater than the male mean. Variance ratios greater than one indicates male variance is greater than female variance. Tail probability ratios greater than 1 indicate higher proportion of boys than girls.

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2.2 standard score points equates to an effect size (d) of 0.15. Following Cohen (1977), in psychological research, 0.2 is considered a small effect, 0.5 is considered medium, and effect sizes above 0.80 are considered large. Too rigid an interpretation of these thresholds can be limiting, because an interpretation of 'large' depends on a number of factors such as the costs of implementing an intervention, the benefits associated with the difference produced, the value attached to the benefits, and so on (see Coe, 2004, for a further discussion). However, it is clear that the effect size for verbal reasoning is very small, and the difference in quantitative and non-verbal reasoning means are negligible.

We can explore sex differences in more detail by considering performance on each of the nine CAT subtests, three within each battery. The mean and standard deviations of raw scores on each Level D subtest were used to calculate the effect sizes, as shown in Table 4. The higher female mean is consistent over all three verbal tests. There appear to be almost no sex difference at all on any of the three quantitative tests (all effect sizes are less than 0.05). For non-verbal reasoning, the overall equality between boys and girls appears to reflect some averaging of a small female advantage in figure classification and a small male advantage in figure analysis. The later test is designed to tap elements of spatial ability on which males often achieve higher scores, although not typically until late adolescence. However, the effect size ($d = -0.09$) is too small to warrant extended discussion.

Table 4. Effect size and variance ratios for gender differences in raw scores on each of the nine CAT Level D subtests

CAT subtest	Effect size (d)	Variance ratio (VR)
Verbal classification	0.16	1.15
Sentence completion	0.12	1.13
Verbal analogies	0.15	1.13
Number analogies	−0.04	1.10
Number series	0.00	1.12
Equation building	0.04	1.16
Figure classification	0.14	1.10
Figure analogies	0.07	1.17
Figure analysis	−0.09	1.07

Note. Positive effect size indicates female mean greater than male mean. VR greater than one indicates male variance greater than female variance.

Sex differences in score variance

The variance ratio is the ratio of male score variance to female score variance. A variance ratio greater than one indicates the variance is greater for boys than for girls, while a variance ratio less than one indicates greater variability in the scores for girls. When viewed as a descriptive statistic, Feingold (1992) suggested that a variance ratio of 1.10 is probably the smallest meaningful effect. In these terms, the sex difference in variability borders the threshold for verbal reasoning, and exceeds this level for both non-verbal and quantitative reasoning (see Table 3). In percentage terms, boys' scores are 9% more variable than girls on verbal reasoning, 13% more variable on non-verbal reasoning, 18% more variable on quantitative reasoning, and 13% more variable on mean CAT score. A similar picture of greater variability in the scores for boys is apparent for all nine of the subtests (Table 4). Male performance is more variable than female, with variance ratios in

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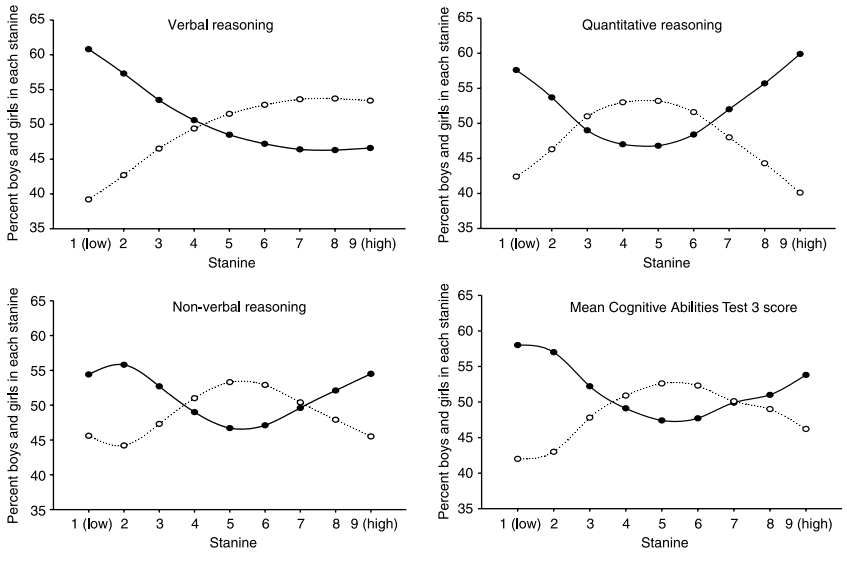

Figure 1. Percentage of boys and girls within each stanine score band for the CAT3's three battery scores and the overall mean CAT3 score. Boys' data are closed circles and girls' data are open circles.

excess of 1.10 on eight of the nine subtests, the exception being figure analysis where the variance ratio was 1.07.

Figure 1 presents a graphical illustration of the percentage of boys and girls within each of nine score bands. These score bands (stanines) split the national distribution into nine bands that approximate the normal curve, as shown in Appendix A. The full data giving participant numbers and percentages are included in Appendix B. Stanines have been selected because this is one of the forms in which CAT scores are routinely reported to test users. The differences in variability are not huge. For example, about 60% of the pupils scoring in the bottom 5% of the VR range, and 60% of those in the top 5% of the QR range, were boys, giving ratios of 1.5:1 and indicating that three of every five pupils identified with these extreme scores will be boys. Differences in the top and bottom 5% of scores for NVR are smaller, with around 55% boys or a ratio of 1.25:1, indicating that five of every nine children identified at these extremes were boys.

For comparison with previous papers, differences in the number of boys/girls with extreme scores are also shown as the ratios of the number of boys to the number of girls who score in the bottom 5%, bottom 10%, top 10% and top 5% of the score distributions (see Table 3). As with the variance ratios, a positive ratio indicates a greater proportion of boys than girls. Males are overrepresented in all extremes, with the exception of the upper tails of the verbal reasoning test. This later result reflects the sex difference in mean score, although it is interesting to note that the underrepresentation of boys in the top 5% would be even larger (0.73:1) were it not for the greater male variance.[1]

[1] *The ratios in Table 3 were empirically determined from the data. However, the test scores were all normally distributed, so the tail proportion ratios could be accurately estimated using the mean and standard deviation for each gender together with the standard normal cumulative distribution function. This allowed modelling of the separate effects of mean and variance differences.*

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Discussion

Sex differences in mean reasoning scores are very small. The only non-trivial sex difference in mean scores was for verbal reasoning, where girls scored on average 2.2 standard age score points above boys. Even here, the effect size was only 0.15, compared with a traditional threshold of 0.20 to infer a small effect size (Cohen, 1977). Comparing the magnitude of the VR–NVR score difference for boys and girls suggests that the sex difference in verbal scores more strongly reflects a relative underperformance by boys in the verbal domain (average VR–NVR difference is − 1.3 points) than an overperformance by girls (average VR–NVR difference is 0.4 points). There were significant differences in the standard deviation of scores between the sexes. Boys' scores were 9%, 13%, and 18% more variable than girls' scores for VR, NVR and QR, respectively. Boys were overrepresented relative to girls in the bottom 5% in verbal reasoning, and at both the bottom and top 5% for quantitative and non-verbal reasoning scores by ratios between 1.2:1 and 1.5:1.

Are the results an artifact of test construction?

It is sometimes argued in relation to sex differences in IQ that the two sexes have been defined rather than discovered to have equal IQ. If test constructors expect equal performance from boys and girls, then they might remove items on which girls show better performance and substitute ones that boost the boys, or vice versa, with the result that both sexes obtain the same mean IQ. However, a review of early studies by Mackintosh suggests that IQ tests were not designed from the outset to yield equal scores for the two sexes, and that early test developers did genuinely discover only small sex differences in mean scores (Mackintosh, 1996, pp. 559–560).

It is true that, guided by the early findings of no significant sex differences, modern IQ and reasoning tests do routinely employ differential item functioning (dif) analyses to reject items with extreme sex differences. However, dif analyses are generally assumed to increase the fairness of the test by removing items where the content is better known by one group than another, and therefore confounds content knowledge and reasoning ability. The dif procedure will eliminate question-specific dif from the test, and may thereby reduce overall score differences, but it will not eliminate any general strength or weakness across all questions, so group differences in overall score will remain. In our view, the absence of substantial sex differences in the mean scores on the CAT is unlikely to be attributable to test construction.

It is difficult to see how test construction issues could account for the observed greater variability in boys' scores. One possibility would be a sex difference in speed-accuracy trade off in a timed test such as the CAT. If boys worked faster but less accurately than girls, then they would be more likely than girls to attempt the harder items at the end of the test, and also less likely to be successful on the easiest items at the start. In such a scenario, mean scores for boys and girls might be identical, but less able boys might obtain a higher proportion of low scores than girls of otherwise similar ability, while more able boys might gain a higher proportion of high scores than girls of otherwise similar ability.

Item data for a nationally representative stratified sample of over 2,000 pupils taking Level D in the summer 2000 UK standardization of CAT3 were examined to test this hypothesis. The data indicate that a speed–accuracy trade off might operate in the QR tests, as 11 of the 14 questions showing significant dif were within five questions of the beginning or end of one or other of the three subtests; the three items favouring girls were located within the first five items of a subtest, and eight of the 11 items favouring

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boys were in the last five of a subtest. This might account for some part of the particularly large sex difference in variance for QR scores. On the other hand, the fact that greater male variability is also reported on the quantitative measures of the WAIS (Feingold, 1992), an untimed, individually administered, graded response test suggests the result is not simply a product of the multiple-choice timed format of CAT. There was also no observable pattern of sex differences related to item difficulty or item position for the VR or NVR tests. Overall, it seems unlikely therefore that the greater variability in boys' scores is simply an artifact of the test.

Differences in mean and variability of scores

In contrast to the widely cited sex differences in mean scores reported by Maccoby and Jacklin (1974), the current results support later studies that suggest that sex differences in mean reasoning scores are small or non-significant (Feingold, 1992; Mackintosh, 1996). The current study focused on schoolchildren aged 11 to 12 and some authors argue that, owing to the faster maturation of girls, sex differences will be obscured up to age 16, appearing only in late adolescence or young adulthood (Lynn, 1994). However, decisions about whether to remain in education are also made at around age 16, and school/college populations become increasingly self-selected, with males more likely to drop out of education, such that results cannot be generalized to the broader population (Hyde *et al.*, 1990, p. 150). Even before this age, there is selection into school subjects that shows sex bias, something that has not occurred at ages 11–12. Studies that have been able to analyse large and nationally representative populations in the 14–17 age range have failed to report large sex differences (Hedge & Nowell, 1995).

In relation to sex differences in variability, the current results support the general finding of greater male variability. The sex difference in variability was least pronounced for verbal reasoning, and most pronounced for quantitative reasoning, congruent with Macoby and Jacklin (1974), Feingold (1992) and Hedges and Nowell (1995). However, the results also show significantly greater male variability for non-verbal reasoning, not previously reported. Most importantly, the current results for quantitative reasoning and non-verbal reasoning show boys simultaneously overrepresented in both the very low and the very high score groups, while Hedges and Nowell show boys overrepresented either at the lower or the upper score range depending on the particular test. Our result is congruent with the other large UK study, which found an excess of boys with both extreme low and high scores (Deary *et al.*, 2003).

Implications of the results

Reasoning scores at age 11 are strongly correlated with subsequent educational attainment in national tests of English, mathematics and science at age 14 in England, and with public examination results at age 16 in England and Scotland; such validity data is particularly strong for the CAT (Strand, 2003; Smith *et al.*, 2001). Given this close association, the lack of substantial sex differences in reasoning scores suggests there is no *a priori* rationale, based on mental ability differences, to expect a large gender gap in subsequent test or examination attainment at age 16. If we wish to look for explanations of the gender gap at GCSE we must look beyond conceptions of ability.

Despite the prominent media and government focus on the gender gap, educators must be careful to avoid general conceptions of boys as under-achievers. It is clear from the current study that boys are slightly more likely to be overrepresented relative to girls at the high as well as the low extremes of reasoning scores. The differential is not great,

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but we might hypothesize that there might be a greater proportion of boys within some of the programmes aimed at addressing the needs of the more able students, such as the gifted and talented strand of the Excellence in Cities programme in England. Of course, the degree of overlap in the score distributions of the sexes is vastly greater than the differences between them, and individual pupils should always be considered on the basis of their actual scores rather than their group membership.

The greater variability in boys' reasoning scores may partly explain their greater representation within populations with special educational needs and among those who fail to achieve any GCSE or equivalent passes (6.4% of boys versus 4.3% of girls aged over 15, a ratio of 1.49:1 (Department for Education and Skills, 2002). However, boys do not appear to be overrepresented at the higher end of GCSE performance. In 2002, only 2.8% of boys' GCSE entries were grade A*, compared with 4.4% of girls entries, a ratio of 0.64:1. Only in economics, mathematics, and physics did boys exceed girls in the proportion of A* grades awarded (Office for Standard in Education, 2002). To this extent, it may be valid to speak of a degree of under-achievement, particularly among more cognitively able boys. It is possible though that sex differences in GCSE examinations reflect wider factors related to motivation and effort, such as girls greater likelihood to complete and submit coursework (Office of Her Majesty's Chief Inspector, 1997), gendered patterns of subject choice (Arnot, David, & Weiner, 1996) or gendered allocation to tiered subjects (Elwood, 1995). Salisbury, Rees, and Gorard (1999) provide a good review of the literature in this area.

Some authors (Heim, 1970; Deary *et al.*, 2003) have also suggested that the differences in variability between the sexes in IQ might account for some of the variability in long-term life outcomes related to cognition, for example the fact that men are slightly more likely to achieve third- and first-class university degrees, and less likely to achieve second-class degrees, than women (Smith & Naylor, 2001). While degree classifications are not simply a product of the subject studied or student's social class (Smith & Naylor, 2001), the many social variables intervening between cognitive abilities at age 11 and later adulthood indicate the need for a complex analysis of such outcomes.

GCSE public examinations rely heavily on essays and other modes of assessment requiring extended writing. We also know the largest sex differences reflect girls' superiority in the area of writing. For example, Cole (1997) reports an analysis of multiple US national datasets for school students assessed at age 9, 14 and 17 on a wide variety of tests. There was no sex difference on verbal reasoning/vocabulary (0.05), a small advantage in verbal reading (0.20), a medium female advantage in verbal language use (0.40), and the largest difference for verbal writing (0.60). If there is a desire to circumvent such sex-based superiority in writing skill, it is important that public examinations continue to utilize a range of assessment methodologies, including non-discursive modes.

Finally, it is worth remembering that the gender gap in performance at all ages, even in GCSE at age 16, is extremely small relative to differences associated with, for example, socio-economic circumstances (e.g. Demack, Drew, & Grimsley, 2000; Strand, 1999). The high media attention given to the gender gap should not distract policy makers from attempting to ameliorate other, more sizable gaps.

Explanations for greater male variability in reasoning scores

The authors of the earliest research suggesting greater male variability championed biological interpretations (Feingold, 1992, p. 63). Evolutionary explanations of greater

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male variability in intellectual abilities continue to abound (e.g. Archer & Mehdikhani, 2003). For a long time, any research reporting sex differences in variability was taken to support the variability hypothesis and automatically assumed to be consistent with an innate explanation. However, findings of sex differences in variability are not inconsistent with cultural or environmental explanations. For example, Hollingworth (1922) argued that men's occupational roles were less constraining than women's, affording men greater diversity in educational and environmental experiences, which could engender greater male variability. Noddings (1992) argued that, 'Girls who remain in school will for the most part listen to the teacher, do at least some assignments, and generally conform sufficiently to avoid landing at the very bottom of any distribution. Similarly, many of the brightest girls still feel pressed not to exhibit or actively enhance their superior test-taking and scoring capabilities' (p. 88). Just as it is widely accepted that differences in reasoning scores are a result of both genetic and environmental influences, so might differences in variability be a consequence of the interaction of such influences.

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Appendix A.

Percentage of pupils in each stanine band

Stanine	National percentage of pupils[a]	Corresponding SAS
9	4	127 and above
8	7	119–126
7	12	112–118
6	17	104–111
5	20	97–103
4	17	89–96
3	12	82–88
2	7	74–81
1	4	73 and below

[a] Percentages have been rounded to the nearest whole number.

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Appendix B.

Numbers and percentages of male and female pupils in each stanine on each CAT3 battery and for mean CAT3 score

		Verbal stanine									
		1	2	3	4	5	6	7	8	9	Total
Boys	*N*	8,455	14,171	17,596	29,308	30,490	27,544	16,037	9,857	4,635	158,093
	%	5.3%	9.0%	11.1%	18.5%	19.3%	17.4%	10.1%	6.2%	2.9%	
Girls	*N*	5,448	10,570	15,312	28,591	32,385	30,830	18,557	11,443	5,321	158,457
	%	3.4%	6.7%	9.7%	18.0%	20.4%	19.5%	11.7%	7.2%	3.4%	
Total	*N*	13,903	24,741	32,908	57,899	62,875	58,374	34,594	21,300	9,956	316,550
	%	4.4%	7.8%	10.4%	18.3%	19.9%	18.4%	10.9%	6.7%	3.1%	
		Quantitative stanine									
		1	2	3	4	5	6	7	8	9	Total
Boys	*N*	3,138	19,634	18,258	29,037	23,255	30,376	16,504	12,565	5,095	157,862
	%	2.0%	12.4%	11.6%	18.4%	14.7%	19.2%	10.5%	8.0%	3.2%	
Girls	*N*	2,313	16,905	19,002	32,707	26,438	32,413	15,215	10,007	3,406	158,406
	%	1.5%	10.7%	12.0%	20.6%	16.7%	20.5%	9.6%	6.3%	2.2%	
Total	*N*	5,451	36,539	37,260	61,744	49,693	62,789	31,719	22,572	8,501	316,268
	%	1.7%	11.6%	11.8%	19.5%	15.7%	19.9%	10.0%	7.1%	2.7%	
		Non-verbal reasoning stanine									
		1	2	3	4	5	6	7	8	9	Total
Boys	*N*	1,390	18,144	20,713	29,245	25,720	27,077	18,095	11,369	6,077	157,830
	%	0.9%	11.5%	13.1%	18.5%	16.3%	17.2%	11.5%	7.2%	3.9%	
Girls	*N*	1,165	14,370	18,564	30,488	29,342	30,458	18,387	10,450	5,075	158,299
	%	0.7%	9.1%	11.7%	19.3%	18.5%	19.2%	11.6%	6.6%	3.2%	
Total	*N*	2,555	32,514	39,277	59,733	55,062	57,535	36,482	21,819	11,152	316,129
	%	0.8%	10.3%	12.4%	18.9%	17.4%	18.2%	11.5%	6.9%	3.5%	
		Mean CAT3 score stanine									
		1	2	3	4	5	6	7	8	9	Total
Boys	*N*	2,505	14,505	19,556	29,917	29,607	30,327	17,960	9,392	2,787	156,556
	%	1.6%	9.3%	12.5%	19.1%	18.9%	19.4%	11.5%	6.0%	1.8%	
Girls	*N*	1,813	10,927	17,872	31,059	32,867	33,269	18,016	9,041	2,394	157,258
	%	1.2%	6.9%	11.4%	19.8%	20.9%	21.2%	11.5%	5.7%	1.5%	
Total	*N*	4,318	25,432	37,428	60,976	62,474	63,596	35,976	18,433	5,181	313,814
	%	1.4%	8.1%	11.9%	19.4%	19.9%	20.3%	11.5%	5.9%	1.7%	